

Algorithmic innovation

two decades from
backpropagation,
then accelerating after 2012;
the Transformer in 2017

Large amounts of data

a by-product of the consumer
internet
and Moore's law for storage

Cheap parallel hardware

gaming GPUs, then chips
designed
for deep learning from the
ground up

A software stack

CUDA, TensorFlow / JAX /
PyTorch
for autodiff, Keras for access

The revolution

slowly at first, and then suddenly

